TEMCO LEARN

Register maps

TStat registers, part 2 · built Nov 13, 2026

TStat registers, part 2

The Modbus register map for the TStat, in address order. Addresses are decimal, and where the map gives a range the register occupies every address in it. This part begins at address 209.0 and ends at 324.0.

Address	Bytes	Register and description
209.0	1.0	LED6 — 8.0 — = — COOLING2 STAGE — 22.0 — = — HEAT3 OR COOL3
210.0	1.0	LED7 — 9.0 — = — COOLING3 STAGE — 23.0 — = — COOL1, 2, OR 3
10.0	=	HEATING1 STAGE — 24.0 — = — HEAT1, 2, OR 3
11.0	=	HEATING2 STAGE — 25.0 — = — OCCUPANCY
12.0	=	HEATING3 STAGE — 26.0 — = — STAGE 1, 2 OR 3
13.0	=	COOLING MODE — 27.0 — = — STAGE 2 OR 3
211.0	1.0	Unoccupied Override Timer, Ort. 0=disabled, >0=number of minutes manual override is allowed
212.0	1.0	OVERRIDE_TIMER_DOWN_COUNT - Number of minutes remaining on the timer when unoccupied override timer is in effect.
213.0	1.0	Temperature sensor filter, FIL, weighted average of stored value to new raw value
214.0	1.0	Heating cooling mode configuration, HC, 0=PID, 1=Keypad, 2=Digital_in1, 3=Digital_in1, 4=Analog_in1, 5=Analog_in2
215.0	2.0	Internal Temperature Sensor IC - Intenral temp chip, not used in the current hardware
216.0	2.0	Internal Thermistor Sensor - Shows the filtered, calibrated value of the internal thermistor sensor
217.0	2.0	Calibration for the internal thermistor - internally managed offset for the internal temp sensor value
218.0	2.0	Calibration for analog input2 - can write here or write to the analog input and this will be calculated automatically
219.0	2.0	Lookup Table 1 - 0.0V value Sensor value that corresponds to 0.0V
220.0	2.0	Lookup Table 1 - 0.5V value Sensor value that corresponds to 0.5V
221.0	2.0	Lookup Table 1 - 1.0V value Sensor value that corresponds to 1.0V
222.0	2.0	Lookup Table 1 - 1.5V value Sensor value that corresponds to 1.5V
223.0	2.0	Lookup Table 1 - 2.0V value Sensor value that corresponds to 2.0V
224.0	2.0	Lookup Table 1 - 2.5V value Sensor value that corresponds to 2.5V
225.0	2.0	Lookup Table 1 - 3.0V value Sensor value that corresponds to 3.0V
226.0	2.0	Lookup Table 1 - 3.5V value Sensor value that corresponds to 3.5V

Address	Bytes	Register and description
227.0	2.0	Lookup Table 1 - 4.0V value Sensor value that corresponds to 4.0V
228.0	2.0	Lookup Table 1 - 4.5V value Sensor value that corresponds to 4.5V
229.0	2.0	Lookup Table 1 - 5.0V value Sensor value that corresponds to 5.0V
230.0	2.0	Lookup Table 2 - 0.0V value Sensor value that corresponds to 0.0V
231.0	2.0	Lookup Table 2 - 0.5V value Sensor value that corresponds to 0.5V
232.0	2.0	Lookup Table 2 - 1.0V value Sensor value that corresponds to 1.0V
233.0	2.0	Lookup Table 2 - 1.5V value Sensor value that corresponds to 1.5V
234.0	2.0	Lookup Table 2 - 2.0V value Sensor value that corresponds to 2.0V
235.0	2.0	Lookup Table 2 - 2.5V value Sensor value that corresponds to 2.5V
236.0	2.0	Lookup Table 2 - 3.0V value Sensor value that corresponds to 3.0V
237.0	2.0	Lookup Table 2 - 3.5V value Sensor value that corresponds to 3.5V
238.0	2.0	Lookup Table 2 - 4.0V value Sensor value that corresponds to 4.0V
239.0	2.0	Lookup Table 2 - 4.5V value Sensor value that corresponds to 4.5V
240.0	2.0	Lookup Table 2 - 5.0V value Sensor value that corresponds to 5.0V
241.0	2.0	Universal PID input select, 0=none, 1=analog_in1, 2=analog_in2
242.0	2.0	Universal PID upper deadband
243.0	2.0	Universal PID lower deadband
244.0	2.0	Universal PID pterm
245.0	2.0	Universal PID iterm
246.0	2.0	Universal PID setpoint
247.0	1.0	Output 1 PID Control 0 = PID1, can assign each output to either PID1 or 2, the max or the min of the two PIDS
248.0	1.0	Output 2 PID Control 1 = PID2
249.0	1.0	Output 3 PID Control 2 = Maximum of PID1 and PID2
250.0	1.0	Output 4 PID Control 3 = Minimum of PID1 and PID2
251.0	1.0	Output 5 PID Control

Address	Bytes	Register and description
252.0	1.0	Output 6 PID Control
253.0	1.0	Output 7 PID Control
254.0	1.0	PID2 Output table- Coasting
255.0	1.0	PID2 Output table- Cooling1
256.0	1.0	PID2 Output table- Cooling2
257.0	1.0	PID2 Output table- Cooling3
258.0	1.0	PID2 Output - Heating1
259.0	1.0	PID2 Output - Heating
260.0	1.0	PID2 Output - Heating3
261.0	1.0	PID2 Valve Output - Coasting
262.0	1.0	Universal PID Valve Output - Cooling1
263.0	1.0	Universal PID Valve Output - Cooling2
264.0	1.0	Universal PID Valve Output - Cooling3
265.0	1.0	Universal PID Valve Output - Heating1
266.0	1.0	Universal PID Valve Output - Heating2
267.0	1.0	Universal PID Valve Output - Heating3
268.0	1.0	Number of Heating Stages in Universal Table-(Maximum # of total heating and cooling states is 6)
269.0	1.0	Number of Cooling Stages in Universal Table-(Maximum # of total heating and cooling states is 6)
270.0	1.0	PID2, the current value of PID number 2
271.0	2.0	PID1 Units High byte - Upper 2 bytes of the PID1 units in ASCII
272.0	2.0	PID1 Units Low byte - Lower 2 bytes of the PID1 units in ASCII
273.0	2.0	PID2 Units High byte - Upper 2 bytes of the PID2 units in ASCII
274.0	2.0	PID2 Units Low byte - Lower 2 bytes of the PID2 units in ASCII
275.0	2.0	PID2 Unoccupied (OFF) Setpoint

Address	Bytes	Register and description
276.0	1.0	Number of Heating Stages in Original Table - (Maximum # of total heating and cooling states is 6)
277.0	1.0	Number of Cooling Stages in Original Table - (Maximum # of total heating and cooling states is 6)
278.0	1.0	PID2 heating or cooling state derived from PID2 register 240, 0=coasting, 1=cooling1, 2=cooling2, 3=cooling3, 4=heating1, 5=heating2, 6=heating3, 14=cooling4, 15=cooling5, 16=cooling6, 17=heating4, 18=heating5, 19=heating6.
279.0	1.0	Valve travel time. The time of the valve travel from one end to another end. The units is second.
280.0	1.0	Determine the output1 mode. Output1 always is ON/OFF mode
281.0	1.0	Determine the output2 mode. Output2 always is ON/OFF mode
282.0	1.0	Determine the output3 mode. Output3 always is ON/OFF mode
283.0	1.0	Determine the output4 mode. 0, ON/OFF mode; 1, floating valve for cooling; 2, lighting control; 3, PWM
284.0	1.0	Determine the output5 mode. 0, ON/OFF mode; 1, floating valve for heating; 2, lighting control; 3, PWM
285.0	1.0	Valve percent. Show the valve opened how much percent. READ ONLY
286.0	1.0	Interlock for output1
287.0	1.0	Interlock for output2
288.0	1.0	Interlock for output3
289.0	1.0	Interlock for output4
290.0	1.0	Interlock for output5
291.0	1.0	Interlock for output6
292.0	1.0	Interlock for output7
293.0	1.0	Setpoint increment on the display each time the user hits the up/down buttons. Units are 0.1Deg, 10 = 1Deg and so on.
294.0	2.0	Last key pressed counter. Minutes since the used last pressed a key.
295.0	1.0	Freeze protect setpoint. If the ambient temperature less than the setpoint, the heating valve will open some time the Delay to off register set .

Address	Bytes	Register and description
296.0	1.0	Delay to open. The heating valve will open if the ambient temp less than the Freeze temp setpoint last the time this register set. The units is second.
297.0	1.0	Delay to close. The duration the heating valve open. The units is minute.
298.0	1.0	Analog input1 function selection. 0, normal; 1, freeze protect sensor input; 2, occupancy sensor input; 3, sweep off mode; 4, clock mode; 5, change over mode. Refer to dI1 on page13.8 = window interlock
299.0	1.0	Analog input2 function selection. 0, normal; 1, freeze protect sensor input; 2, occupancy sensor input; 3, sweep off mode; 4, clock mode; 5, change over mode. Refer to dI1 on page13.8 =window interlcok
300.0	1.0	dI1 – Digital input 1 function. Refer to dI1 description in Table 1: Advanced Menu Items
301.0	2.0	Period timer ON time.
302.0	2.0	Period timer OFF time.
303.0	1.0	Period timer units. 0, second; 1, minute; 2, hour.
304.0	1.0	Keypad encoded value. Last keypress, READ ONLY
305.0	1.0	LED hundred's digit, , can drive the LEDs manually when the display register 203 is set to manual (5)
306.0	1.0	LED ten's digit code, can drive the LEDs manually when the display register 203 is set to manual (5)
307.0	1.0	LED one's digit code, can drive the LEDs manually when the display register 203 is set to manual (5)
308.0	1.0	LED discrete status symbols, can drive the LEDs manually when the display register 203 is set to manual (5) — Register 309 for TStat5E
309.0	1.0	Input auto/ manual enable. Bit0 correspond to analog input1(register 180); bit1 to analog input2(register 181); bit2 to digital input1(register 311),bit3 correspond to itternal sensor(register 216).0, auto mode, the corresponding input value from sensor; 1, manual mode, the corresponding value from serial port. — Input auto/ manual enable. Bit0 correspond to analog input1(register 367); bit1 to analog input2(register 368); bit7 to analog input8(register 374),bit8 correspond to itternal sensor(register 216).0, auto mode, the corresponding input value
310.0	1.0	Output auto/manual enable. Bit 0 to 4 correspond to output1 to output5, bit 5 correspond to output6(register 102), bit 6 correspond to output7(register 103). 0, auto mode; 1, manual mode. — from sensor; 1, manual mode, the corresponding value from serial port.

Address	Bytes	Register and description
311.0	1.0	Digital manual input. Write the manual value for digital input when digital input in manual mode.
312.0	1.0	(REV35_0) DEADMASTER_AUTO_MANUAL = 0, the default, outputs will not change, stay in whatever mode they were last commanded DEADMASTER_AUTO_MANUAL = 1, the output will be trigger to “AUTO” mode DEADMASTER_AUTO_MANUAL = 2, the outputs will go to manual on or off as defined in register 313.
313.0	1.0	REV 35_0 (relay1-5) output manual value when DEADMASTER_AUTO_MANUAL = 2 register 313, Bit0 is for relay 1, bit1 for relay 2 and so on up to output 5
314.0	2.0	REV 35_0 (analog output - cooling)deadmaster config value during deadmaster be triggered when DEADMASTER_AUTO_MANUAL = 2 Value Range: 0 to 1000
315.0	2.0	REV 35_0 (analog output - heating)deadmaster config value during deadmaster be triggered when DEADMASTER_AUTO_MANUAL = 2 Value Range: 0 to 1000
316.0	1.0	RESERVE
317.0	1.0	Dead master. The Tstat will go to occupied mode automatically if there is no serial communications for a user defined period of time, for example if the register is set to 10 the thermostat will go to ocupied mode if there are no communications for a period of 10 minutes. Set the register to 0 to disable the feature
318.0	1.0	Rouding display. 0 = round the display to the nearest digit; 1 = round the display to the nearest 1/10 unit; 5 = round the display to the nearest 1/2 unit. 2,3,4 reserved.
319.0	1.0	The minimum address which the device can accept, use this to limit addresses to a certain defined band.
320.0	1.0	The maximum device address can be set. The device address should between min and max address, default is 254
321.0	1.0	Rotation features [Old disabled feature] Output 2 is controlled by which output table in the rotation group. READ ONLY.
322.0	1.0	Rotation feature, [disabled]: output3
323.0	1.0	Rotation feature, [disabled]: output4
324.0	1.0	Rotation feature, [disabled]: output5